

SSHVisual: Visual Tool to Analyze SSH Data

Shreeman Gautam*, Robert Ricci*

*University of Utah

Abstract

Network operators use visualizations to detect attacks on their systems, exploring and interacting with data to uncover patterns that would otherwise stay hidden. Operators then use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and uncovering deeper patterns iteratively in data is at best, tedious and at worst, not achievable without data visualizations. Previous works in network visualization analyzed the IP and AS dimensions of SSH logins without much focus on the username aspect of the attackers' attempts, detected anomalous network events only where baseline events were available, and enabled operators to share findings with one another. Little attention has been paid to whether attackers observed at different times represent the same persistent actor. We present an iterative visualization SSHVisual, using login attempts collected on our own SSH-facing systems. SSHVisual analyzes the IP, AS, and username dimensions of SSH attacks, in service of SSH campaign detection. It complements automated detection by allowing operators to find anomalous events without normal baseline events. It correlates SSH attacks with real-world events, and it introduces a similarity metric that quantifies attacker persistence over time. Like previous systems, SSHVisual supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual using case studies and found that it helps us block attacks and uncover patterns. We identified campaigns in which a set of attempted usernames recurred across unrelated IPs and ASes. The similarity metric surfaced attackers returning across different time periods despite changing network origin. SSHVisual shows that treating usernames as a first-class analytic dimension reveals attacker coordination that IP-centric analysis misses, and that exploratory, operator-driven detection remains valuable.

I Introduction

Computer systems worldwide employ SSH [1] to allow their users to securely connect to them in order to execute commands and do remote administration. A lot of users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial in nature, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess what the username, password combinations are and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3, 4] or steal resources from the

systems, like cryptomining [5, 6].

Given the prevalence of SSH attacks in the IPv4 space [2], we collect SSH login data on our own systems and present SSHVisual. SSHVisual serves two kinds of network operators. Network administrators, who handle day-to-day SSH traffic, need to identify attackers quickly and block them. Security analysts, who study traffic over longer time periods, look for patterns that are not immediately apparent and share what they find across organizations. Consider a single time period: an administrator can isolate the highest-volume attacker and block it immediately, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated botnet.

Surveys of network visualization [7, 8, 9, 10, 11] establish the maturity of the field and the gaps we target. Our contributions follow from treating the username as an attacker identity. First, actor-level analysis: we treat the attacker as a first-class actor — including the username identity, not only the source IP or autonomous system (AS) — and we identify coordinated campaigns. Since network data affords no normal baseline for automated detection [12], this identification is necessarily exploratory (therefore, not antagonistic but complementary to automated detection with normal baselines) — operators recognize coordination visually rather than specifying it in advance. Second, a similarity metric that quantifies attacker similarity and lets operators track how actors evolve in the username dimension across time periods. Third, correlation of actor behavior with real-world events, which is more tractable through usernames than IPs, given the limitations of IP corroboration. Additionally, following established practices, we allow for iterative operator driven collaboration and labeling of irregularities found in the data.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion and Future Work.

II Related Work

A long line of tools [13, 14, 15, 16, 17, 18, 19, 20] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally, a smaller alert- and log-oriented group [21, 22, 23, 24, 25, 26, 27, 28, 29, 30] organizes IDS output and heterogeneous logs of which [23] contributes a visual analysis (of the CICIDS2017 dataset) that isolates at-

tack clusters — including SSH brute-force (SSH-Patator) — through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A Attack Campaigns with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track — the unit that, alongside the IP and AS, anchors our analysis. Some papers [31, 32, 33, 34, 35, 36, 37] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted — the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36] — usernames and passwords appear only as static top-N tallies (kippo-graph’s top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time. The WAN honeypot framework [38] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist — a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B Attacker Similarity with Time

Computed similarity between entities is well established in literature [39, 40, 41, 42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources — not external attacker identities [40]. None treats the username as an attacker actor. On time: [41, 42] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics — for example, Jaccard overlap of attackers between successive windows — particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C Limits of automated detection

One premise of our approach is that SSH honeypot traffic does not admit the normal baseline that automated anomaly and campaign detection presuppose. The broader literature already documents the fragility of such detection: Sommer and Paxson [43] note that machine-learning anomaly detection is rarely deployed for intrusion detection given the variability of “normal” the cost of errors, and the difficulty of sound evaluation, while Axelsson [44] shows it is throttled by the base-rate fallacy, where the false-positive rate becomes the limiting factor once intrusions are rare relative to benign traffic. Our setting departs from that one entirely, in the direction Gates and Taylor [12] anticipate when they question whether attacks are rare or a stable normal exists, note that normal traffic may itself

be malicious and malicious traffic unremarkable, and observe that detection ultimately devolves to an operator investigating clusters of activity by hand. The consequence is visible in the systems cited above: Situ [39] learns a probabilistic model of each IP’s behavior and CLIQUE [45] scores each actor against a baseline of its own expected activity, so both presume a per-actor “normal” that near-uniformly hostile, constantly churning traffic cannot supply. Situ nonetheless concedes that automation alone is insufficient — operators “do not want another automated tool with algorithms they do not trust” — and builds a human-in-the-loop system; we reach the same conclusion from a starker starting point, lacking even the per-actor baseline it assumes, and so treat campaign identification as operator-driven from the outset.

D Actor identity and real-world event correlation

The attacks we observe do not arrive at a steady rate; they rise, fall, and change in character as events unfold in the wider world, so understanding them means lining them up against what is happening outside the network. The link is well documented: Bilge and Dumitras [46] found that the number of attacks against a given vulnerability can jump by up to five orders of magnitude — a hundred-thousand-fold increase — once it is publicly disclosed, and Sabottke et al. [47] showed that future exploitation can be predicted from signals in disclosure channels alone. Allodi and Massacci [48] add that a vulnerability’s official severity score is a poor guide to whether it is actually attacked, so the dependable way to learn what attackers are reacting to is to watch real activity and line it up against real events, not to read it off a score. We want to do this at the level of a campaign — a group of attackers acting together — rather than one attacker at a time, and that raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine — one host is assigned many addresses over time [49], hosts cycle through new addresses [50], and counting by address can overstate the number of real machines by orders of magnitude [51] — so it is likely that a view organized by address scatters one campaign across dozens of entries that look unrelated. The username pulls those pieces back together: the same login names likely recur across the changing addresses, so grouping by username re-assembles the campaign that an address-by-address view splits apart. A username is not a perfect identity either — names like “root” are tried by almost everyone, and one campaign runs through a whole list of names — so we treat it as one analysis dimension alongside the address and the network it belongs to (its AS), resolving the overlaps with the similarity and grouping methods above.

E Operator labeling and collaboration

Since detection on our data is operator-driven, the findings an operator produces by hand must be captured and shared rather than discarded. In two lines of work [52, 19], interactive labeling has been used to fold the operator’s judgments back into

the system. Collaborative analysis has been supported through shared artifacts with [17] and [20] being notable examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III Tasks and Requirements

The design of SSHVisual derives from our(authors) experiences as network operators and as mentioned, the gaps and established practices that we have found in network literature. Accordingly, we formulate the tasks and requirements to accommodate network operator needs, fill out gaps and respect established practices.

The visual tasks for network operators are as follows: Find SSH campaigns, using data that has Username, IP, AS, and other information, and publish them/share that information(T1); Find anomalies from data, block them and publish them/share that information(T2); Allow iterative collaboration based on visualized information(T3). T1 is aimed primarily at security analysts, T2 is aimed primarily at network administrators and T3 is aimed at both analysts and administrators.

The first requirement (R1) is profiling attacker similarity. The visualization should be able to algorithmically cluster similar attackers based primarily along the username dimension, secondarily along IP and AS information and tertiarily along other dimensions. R1 mostly serves T1 as it allows analysts to build a repository of similar attackers.

The second requirement (R2) is linking data events to real life events. The visualization should be able to show, if any, news artifacts that correspond similar information as observed in data. If there is nothing to show, that is valuable new information that the operator has uncovered and which others can use if the operator shares that information. R2 serves both T1 and T2 as it allows analysts and administrators to either build on existing attackers knowledge base or share new attackers to interested parties. Keep in mind that the visualization itself does not allow the operators to publish info; we assume that the operators can share their knowledge using typed up reports and screenshots in existing network monitoring websites.

The third requirement (R3) is filtering of variables. The visualization should be able to show attackers who fit constraint(s) on variables as specified by the operator. R3 primarily fulfills T1 and T2 because administrators can filter out anomalies and analysts can find similar attackers given constraints.

The fourth requirement (R4) is minimal visual overload. Line charts show 10 attackers at most (unless the operator wants to see more) using username, IP and AS along a temporal dimension. At the same time, an aggregate graph shows all the attacks without specifications. For instance, if 150M attacks happened on a given day in the aggregate graph and 75M of that came from one username/IP/AS in the username/IP/AS graph, those attackers can be identified easily. R4 primarily fulfills T2 as it allows for easier detection and eventual block-

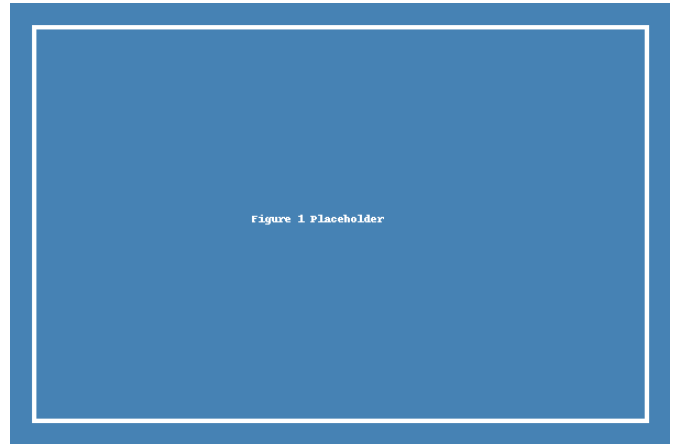


Figure 1: Show campaign detection using username+IP+AS

age of attackers.

The fifth requirement (R5) is tagging/flagging. Any aberrant attacker can be tagged/flagged by the network operator so that operators on the same network do not have to identify the same patterns/anomalies twice. R5 supports T3 as it allows for collaboration.

The sixth requirement (R6) is exploratory workflows and usage of different visualization types. The visual system should allow for filtering and clustering using tabular visualization, linechart visualization based on clustering/non-clustering and filtering/non-filtering, and checking information with real life events. For instance, an administrator can find an anomaly easily using filtering in the tabular visualization and may not have to see that in a linechart to block attackers. However, an analyst may need to use the tabular visualization to build a repository of similar attackers and use the linechart visualization to see those attackers visually. R6 supports T1 and T2 because two different people can use the same system for different means and to achieve different results.

IV Design/Approach – How requirements are fulfilled in the visualization – screenshots go here

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo. Our clustering of daily time series builds on the calendar-based approach of [?], while the multi-dimensional filtering view is inspired by parallel sets [?] and parallel coordinates [?].

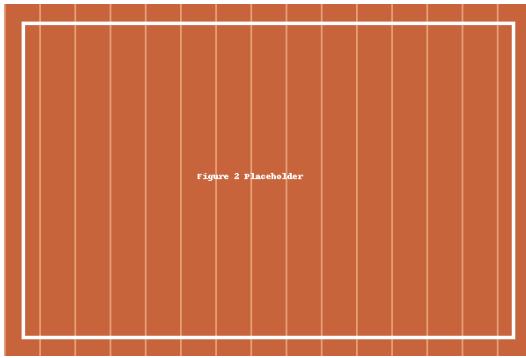


Figure 2: Show collaboration + labeling

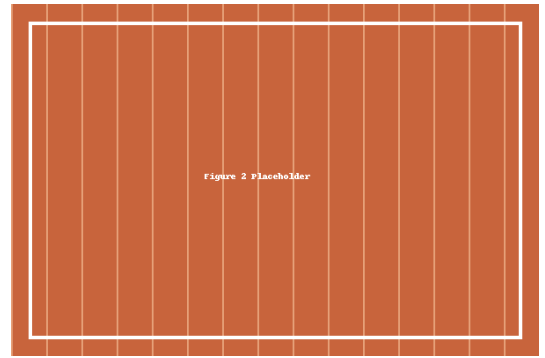


Figure 3: Show similarity metric at play

A First Component

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

B Second Component

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor.

C Third Component

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in, cursus faucibus, augue.

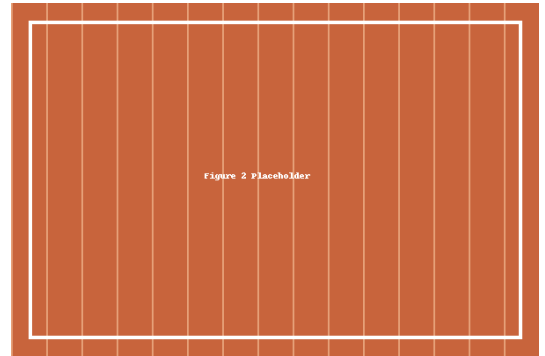


Figure 4: Show correlation of attacks with real events

V Implementation – how we made the tool. this section may not exist by the end

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra. Nullam varius. Etiam dignissim elementum metus. Vestibulum faucibus, metus sit amet mattis rhoncus, sapien dui laoreet odio, nec ultricies nibh augue a enim. Fusce in ligula. Quisque at magna et nulla commodo consequat. Proin accumsan imperdiet sem. Nunc porta. Donec feugiat mi at justo. Phasellus facilisis ipsum quis ante. In ac elit eget ipsum pharetra faucibus. Maecenas viverra nulla in massa.

VI Evaluation and Discussion

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus. Phasellus eu quam vitae turpis viverra pellentesque. Duis feugiat felis ut enim. Phasellus pharetra, sem id porttitor sodales, magna nunc aliquet nibh, nec blandit nisl mauris at pede. Suspendisse risus risus, lobortis eget, semper at, imperdiet sit amet, quam. Quisque scelerisque dapibus nibh. Nam enim. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ut metus. Ut metus justo, auctor at, ultrices eu, sagittis ut, purus. Aliquam aliquam.

Etiam pede massa, dapibus vitae, rhoncus in, placerat po-

suere, odio. Vestibulum luctus commodo lacus. Morbi lacus dui, tempor sed, euismod eget, condimentum at, tortor. Phasellus aliquet odio ac lacus tempor faucibus. Praesent sed sem. Praesent iaculis. Cras rhoncus tellus sed justo ullamcorper sagittis. Donec quis orci. Sed ut tortor quis tellus euismod tincidunt. Suspendisse congue nisl eu elit. Aliquam tortor diam, tempus id, tristique eget, sodales vel, nulla. Praesent tellus mi, condimentum sed, viverra at, consectetur quis, lectus. In auctor vehicula orci. Sed pede sapien, euismod in, suscipit in, pharetra placerat, metus. Vivamus commodo dui non odio. Donec et felis.

A Case Studies

Etiam suscipit aliquam arcu. Aliquam sit amet est ac purus bibendum congue. Sed in eros. Morbi non orci. Pellentesque mattis lacinia elit. Fusce molestie velit in ligula. Nullam et orci vitae nibh vulputate auctor. Aliquam eget purus. Nulla auctor wisi sed ipsum. Morbi porttitor tellus ac enim. Fusce ornare. Proin ipsum enim, tincidunt in, ornare venenatis, molestie a, augue. Donec vel pede in lacus sagittis porta. Sed hendrerit ipsum quis nisl. Suspendisse quis massa ac nibh pretium cursus. Sed sodales. Nam eu neque quis pede dignissim ornare. Maecenas eu purus ac urna tincidunt congue.

B Discussion

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit.

VII Summary

Maecenas non massa. Vestibulum pharetra nulla at lorem. Duis quis quam id lacus dapibus interdum. Nulla lorem. Donec ut ante quis dolor bibendum condimentum. Etiam egestas tortor vitae lacus. Praesent cursus. Mauris bibendum pede at elit. Morbi et felis a lectus interdum facilisis. Sed suscipit gravida turpis. Nulla at lectus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Praesent nonummy luctus nibh. Proin turpis nunc, congue eu, egestas ut, fringilla at, tellus. In hac habitasse platea dictumst.

A Limitations

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum. Suspendisse condimentum, tortor at egestas posuere, neque metus tempor orci, et tincidunt urna nunc a purus. Sed facilisis blandit tellus. Nunc risus sem, suscipit nec, eleifend quis, cursus

quis, libero. Curabitur et dolor. Sed vitae sem. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Maecenas ante. Duis ullamcorper enim. Donec tristique enim eu leo. Nullam molestie elit eu dolor. Nullam bibendum, turpis vitae tristique gravida, quam sapien tempor lectus, quis pretium tellus purus ac quam. Nulla facilisi.

B Future Work

Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim. Aliquam erat volutpat. Proin sit amet nulla ut eros consectetur cursus. Phasellus dapibus aliquam justo. Nunc laoreet. Donec consequat placerat magna. Duis pretium tincidunt justo. Sed sollicitudin vestibulum quam. Nam quis ligula. Vivamus at metus. Etiam imperdiet imperdiet pede. Aenean turpis. Fusce augue velit, scelerisque sollicitudin, dictum vitae, tempor et, pede. Donec wisi sapien, feugiat in, fermentum ut, sollicitudin adipiscing, metus.

Acknowledgments

Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio. Vivamus metus lacus, varius quis, imperdiet quis, rhoncus a, turpis. Etiam ligula arcu, elementum a, venenatis quis, sollicitudin sed, metus. Donec nunc pede, tincidunt in, venenatis vitae, faucibus vel, nibh. Pellentesque wisi. Nullam malesuada. Morbi ut tellus ut pede tincidunt porta. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam congue neque id dolor.

References

- [1] T. Ylonen, "Ssh-secure login connections over the internet," in *Proceedings of the 6th USENIX Security Symposium*, vol. 37, pp. 40–52, 1996.
- [2] S. K. Singh, S. Gautam, C. Cartier, S. Patil, and R. Ricci, "Where the wild things are: Brute-Force SSH attacks in the wild and how to stop them," in *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*, (Santa Clara, CA), pp. 1731–1750, USENIX Association, Apr. 2024.
- [3] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis, *et al.*, "Understanding the mirai botnet," in *26th USENIX security symposium (USENIX Security 17)*, pp. 1093–1110, 2017.
- [4] C. Koliass, G. Kambourakis, A. Stavrou, and J. Voas, "Ddos in the iot: Mirai and other botnets," *computer*, vol. 50, no. 7, pp. 80–84, 2017.
- [5] F. Dang, Z. Li, Y. Liu, E. Zhai, Q. A. Chen, T. Xu, Y. Chen, and J. Yang, "Understanding fileless attacks on

- linux-based iot devices with honeycloud,” in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, pp. 482–493, 2019.
- [6] S. Pastrana and G. Suarez-Tangil, “A first look at the crypto-mining malware ecosystem: A decade of unrestricted wealth,” in *Proceedings of the Internet Measurement Conference*, pp. 73–86, 2019.
 - [7] V. T. Guimarães, C. M. D. S. Freitas, R. Sadre, L. M. R. Tarouco, and L. Z. Granville, “A survey on information visualization for network and service management,” *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 285–323, 2016.
 - [8] H. Shiravi, A. Shiravi, and A. A. Ghorbani, “A survey of visualization systems for network security,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 8, pp. 1313–1329, 2012.
 - [9] T. Zhang, X. Wang, Z. Li, F. Guo, Y. Ma, and W. Chen, “A survey of network anomaly visualization,” *Science China Information Sciences*, vol. 60, no. 12, p. 121101, 2017.
 - [10] J. Raynor, T. Crnovrsanin, S. Di Bartolomeo, L. South, D. Saffo, and C. Dunne, “The state of the art in bgp visualization tools: A mapping of visualization techniques to cyberattack types,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 1059–1069, 2022.
 - [11] C. Wu, S. Sheng, and X. Dong, “Research on visualization systems for ddos attack detection,” in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 2986–2991, IEEE, 2018.
 - [12] C. Gates and C. Taylor, “Challenging the anomaly detection paradigm: A provocative discussion,” in *Proceedings of the 2006 workshop on New security paradigms*, pp. 21–29, 2006.
 - [13] F. Mansmann and S. Mansmann, “Interactive exploration of data traffic with hierarchical network maps,” 2006.
 - [14] J. Oberheide, M. Goff, and M. Karir, “Flamingo: Visualizing internet traffic,” in *2006 IEEE/IFIP Network Operations and Management Symposium NOMS 2006*, pp. 150–161, IEEE, 2006.
 - [15] F. Mansmann, F. Fischer, D. A. Keim, and S. C. North, “Visual support for analyzing network traffic and intrusion detection events using treemap and graph representations,” in *Proceedings of the Symposium on Computer Human Interaction for the Management of Information Technology*, pp. 19–28, 2009.
 - [16] T. Taylor, D. Paterson, J. Glanfield, C. Gates, S. Brooks, and J. McHugh, “Flovis: Flow visualization system,” in *2009 Cybersecurity Applications & Technology Conference for Homeland Security*, pp. 186–198, IEEE, 2009.
 - [17] S. Chen, C. Guo, X. Yuan, F. Merkle, H. Schaefer, and T. Ertl, “Oceans: Online collaborative explorative analysis on network security,” in *Proceedings of the Eleventh Workshop on Visualization for Cyber Security*, pp. 1–8, 2014.
 - [18] T. Tremel, J. Kogel, F. Jauernig, S. Meier, D. Thom, F. Becker, C. Muller, and S. Koch, “Vitalflow: Visual interactive traffic analysis with netflow,” in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*, pp. 1–6, IEEE, 2022.
 - [19] L. Xiao, J. Gerth, and P. Hanrahan, “Enhancing visual analysis of network traffic using a knowledge representation,” in *2006 IEEE symposium on visual analytics science and technology*, pp. 107–114, IEEE, 2006.
 - [20] J. R. Goodall and M. Sowul, “Viassist: Visual analytics for cyber defense,” in *2009 IEEE conference on technologies for homeland security*, pp. 143–150, IEEE, 2009.
 - [21] Y. Livnat, J. Agutter, S. Moon, and S. Foresti, “Visual correlation for situational awareness,” in *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.*, pp. 95–102, IEEE, 2005.
 - [22] A. King, B. Huffaker, A. Dainotti, and K. Claffy, “A coordinated view of the temporal evolution of large-scale internet events,” *Computing*, vol. 96, no. 1, pp. 53–65, 2014.
 - [23] D. H. Jeong, B.-K. Jeong, and S.-Y. Ji, “Multi-resolution analysis with visualization to determine network attack patterns,” *Applied Sciences*, vol. 13, no. 6, p. 3792, 2023.
 - [24] D. H. Jeong, J.-H. Cho, F. Chen, L. Kaplan, A. Jøsang, and S.-Y. Ji, “Interactive web-based visual analysis on network traffic data,” *Information*, vol. 14, no. 1, p. 16, 2022.
 - [25] D. Ceneda, M. Di Bartolomeo, V. Di Donato, M. Patrignani, M. Pizzonia, and M. Rimondini, “Routingwatch: Visual exploration and analysis of routing events,” in *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*, pp. 591–597, IEEE, 2016.
 - [26] J. C. Roberts, “State of the art: Coordinated & multiple views in exploratory visualization,” in *Fifth international conference on coordinated and multiple views in exploratory visualization (CMV 2007)*, pp. 61–71, IEEE, 2007.
 - [27] J. S. Yi, Y.-a. Kang, J. T. Stasko, and J. A. Jacko, “Understanding and characterizing insights: how do people gain insights using information visualization?,” in *Proceedings of the 2008 Workshop on BEyond time and errors: novel evaluation methods for Information Visualization*, pp. 1–6, 2008.

- [28] H. Choi, H. Lee, and H. Kim, "Fast detection and visualization of network attacks on parallel coordinates," *computers & security*, vol. 28, no. 5, pp. 276–288, 2009.
- [29] C. Humphries, N. Prigent, C. Bidan, and F. Majorczyk, "Elvis: Extensible log visualization," in *Proceedings of the Tenth Workshop on Visualization for Cyber Security*, pp. 9–16, 2013.
- [30] T. Yu, R. Lippmann, J. Riordan, and S. Boyer, "Ember: a global perspective on extreme malicious behavior," in *Proceedings of the seventh international symposium on visualization for cyber security*, pp. 1–12, 2010.
- [31] R. Djap, C. Lim, K. E. Silaen, and A. Yusuf, "Xb-pot: Revealing honeypot-based attacker's behaviors," in *2021 9th International Conference on Information and Communication Technology (ICoICT)*, pp. 550–555, IEEE, 2021.
- [32] V. Visoottiviseth, U. Jaralrungrroj, E. Phoomrungraungsuk, and P. Kultanon, "Distributed honeypot log management and visualization of attacker geographical distribution," in *2011 Eighth International Joint Conference on Computer Science and Software Engineering (JCSSE)*, pp. 23–28, IEEE, 2011.
- [33] M. Baykara and R. Das, "A novel honeypot based security approach for real-time intrusion detection and prevention systems," *Journal of Information Security and Applications*, vol. 41, pp. 103–116, 2018.
- [34] A. Belqruch and A. Maach, "Scada security using ssh honeypot," in *Proceedings of the 2nd International Conference on Networking, Information Systems & Security*, pp. 1–5, 2019.
- [35] I. Koniaris, G. Papadimitriou, and P. Nicopolitidis, "Analysis and visualization of ssh attacks using honeypots," in *Eurocon 2013*, pp. 65–72, IEEE, 2013.
- [36] A. Kyriakou and N. Sklavos, "Container-based honeypot deployment for the analysis of malicious activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*, pp. 1–4, IEEE, 2018.
- [37] I. Koniaris, G. Papadimitriou, P. Nicopolitidis, and M. Obaidat, "Honeypots deployment for the analysis and visualization of malware activity and malicious connections," in *2014 IEEE international conference on communications (ICC)*, pp. 1819–1824, IEEE, 2014.
- [38] D. Fraunholz, M. Zimmermann, S. D. Anton, J. Schneider, and H. D. Schotten, "Distributed and highly-scalable wan network attack sensing and sophisticated analysing framework based on honeypot technology," in *2017 7th International Conference on Cloud Computing, Data Science & Engineering-Confluence*, pp. 416–421, IEEE, 2017.
- [39] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 204–214, 2018.
- [40] Q. Liao and A. Striegel, "Intelligent network management using graph differential anomaly visualization," in *2012 IEEE Network Operations and Management Symposium*, pp. 1008–1014, IEEE, 2012.
- [41] W. A. Pike, C. Scherrer, and S. Zabriskie, "Putting security in context: Visual correlation of network activity with real-world information," in *VizSEC 2007: Proceedings of the Workshop on Visualization for Computer Security*, pp. 203–220, Springer, 2008.
- [42] L. Hao, C. G. Healey, and S. E. Hutchinson, "Ensemble visualization for cyber situation awareness of network security data," in *2015 IEEE Symposium on Visualization for Cyber Security (VizSec)*, pp. 1–8, IEEE, 2015.
- [43] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *2010 IEEE symposium on security and privacy*, pp. 305–316, IEEE, 2010.
- [44] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security (TISSEC)*, vol. 3, no. 3, pp. 186–205, 2000.
- [45] D. M. Best, R. P. Hafen, B. K. Olsen, and W. A. Pike, "Atypical behavior identification in large-scale network traffic," in *2011 IEEE Symposium on Large Data Analysis and Visualization*, pp. 15–22, IEEE, 2011.
- [46] L. Bilge and T. Dumitras, "Before we knew it: an empirical study of zero-day attacks in the real world," in *Proceedings of the 2012 ACM conference on Computer and communications security*, pp. 833–844, 2012.
- [47] C. Sabottke, O. Suciu, and T. Dumitras, "Vulnerability disclosure in the age of social media: Exploiting twitter for predicting real-world exploits," in *24th USENIX security symposium (USENIX security 15)*, pp. 1041–1056, 2015.
- [48] L. Allodi and F. Massacci, "A preliminary analysis of vulnerability scores for attacks in wild: The ekits and sym datasets," in *Proceedings of the 2012 ACM Workshop on Building analysis datasets and gathering experience returns for security*, pp. 17–24, 2012.
- [49] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are ip addresses?," in *Proceedings of the 2007 conference on Applications, technologies, architectures, and protocols for computer communications*, pp. 301–312, 2007.

- [50] B. Stone-Gross, M. Cova, L. Cavallaro, B. Gilbert, M. Szydlowski, R. Kemmerer, C. Kruegel, and G. Vigna, “Your botnet is my botnet: analysis of a botnet takeover,” in *Proceedings of the 16th ACM conference on Computer and communications security*, pp. 635–647, 2009.
- [51] G. C. Moura, C. Ganán, Q. Lone, P. Poursaied, H. Asghari, and M. van Eeten, “How dynamic is the isps address space? towards internet-wide dhcp churn estimation,” in *2015 IFIP Networking Conference (IFIP Networking)*, pp. 1–9, IEEE, 2015.
- [52] J. L. Guerra, E. Veas, and C. A. Catania, “A study on labeling network hostile behavior with intelligent interactive tools,” in *2019 IEEE Symposium on Visualization for Cyber Security (VizSec)*, pp. 1–10, IEEE, 2019.